

Tecnológico de Monterrey

Q-Learning Algorithm

Daniel Otero Fadul

Department of Science
School of Engineering and Science

What is Reinforcement Learning?

Reinforcement learning focuses on discovering the best actions to take in given situations to achieve the highest possible numerical reward. Instead of following explicit instructions, the learner finds high-yielding choices through trial-and-error experimentation.

Because current decisions alter future conditions, they influence not just immediate outcomes but long-term gains, which is why trial-and-error search and delayed rewards are the defining hallmarks of reinforcement learning.

Types of Machine Learning

Supervised

Learns from static, labeled ground-truth targets (x, y).

- Classification
- Regression

Unsupervised

Discovers patterns and latent structure in unlabeled data.

- Clustering
- Dimensionality Reduction

Reinforcement

Optimizes action policy via continuous environment interaction.

- Trial-and-error
- Scalar reward signals

Markov Decision Processes (MDPs)

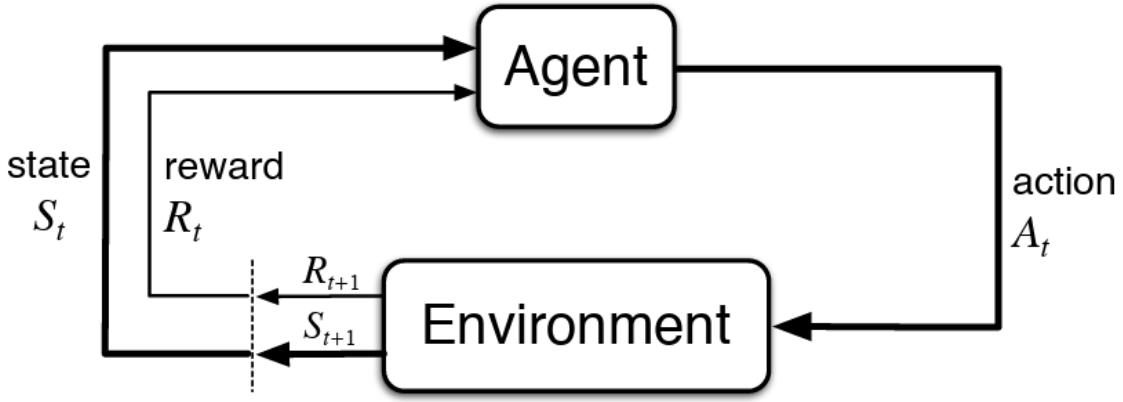


Figura: The agent-enviroment interaction in a Markov decision process. Image taken from [1].

Formal MDP Tuple $\mathcal{M} = \langle S, A, P, R, \gamma \rangle$

- S: Set of all valid states.
- A: Set of all valid actions.
- $P(s' \mid s, a)$: State transition probability distribution.
- $R(s, a, s')$: Reward function.
- $\gamma \in [0, 1]$: Discount factor for future rewards.

Overview of Reinforcement Learning Methods

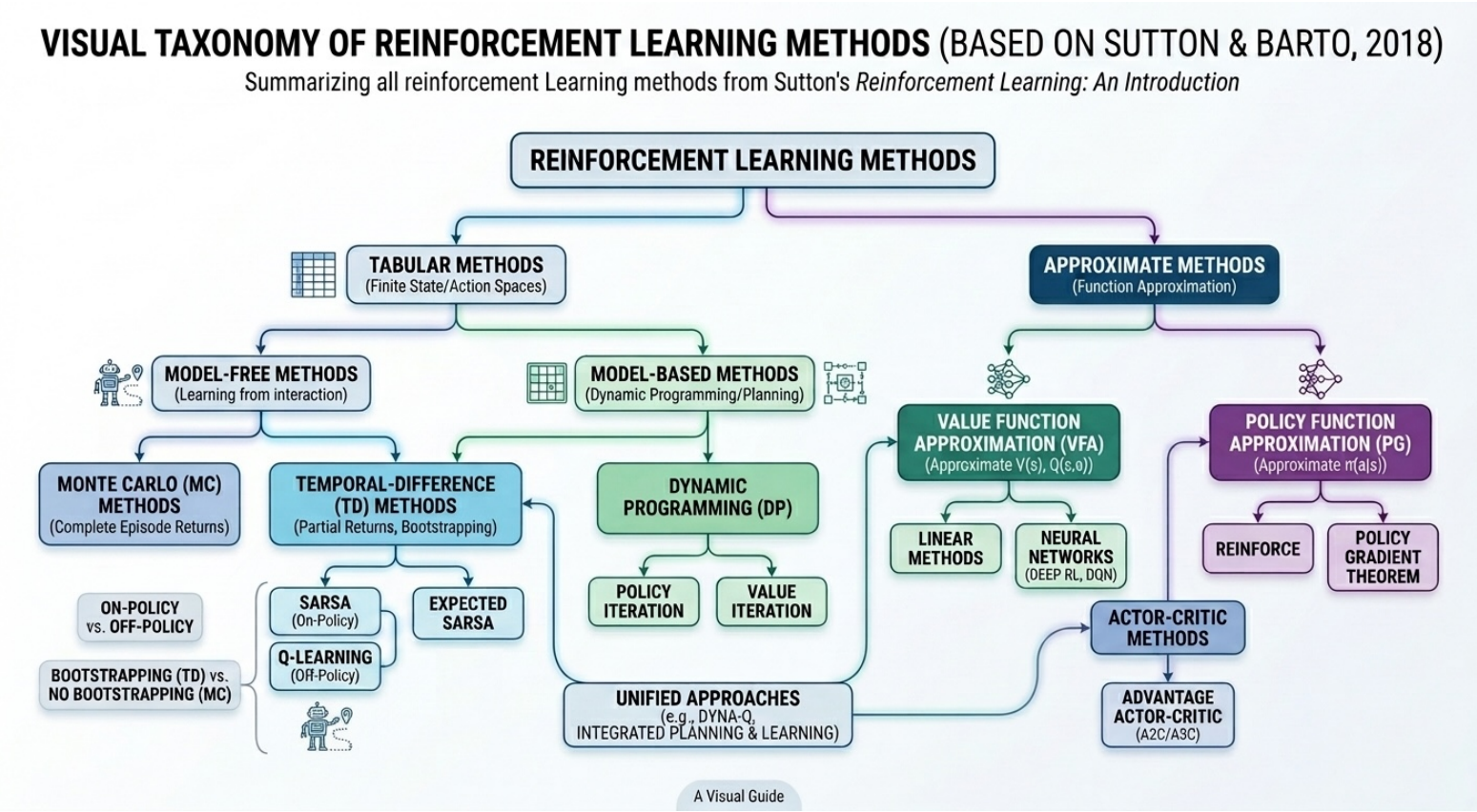


Figura: Visual summary of the methods described in "Reinforcement Learning: An Introduction."

What is Q-Learning?

Q-learning is a widely used model-free reinforcement learning technique that enables an agent to make optimal decisions through direct interaction with its surroundings. Rather than relying on a prebuilt model of the environment, the agent learns purely through hands-on trial and error by testing different actions and evaluating their outcomes.

Q-Learning uses the **action-value function** $Q(s, a)$, which estimates the expected cumulative future discounted rewards for taking action a in state s and thereafter following a policy π:

$$Q_\pi(s, a) = \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \,\middle|\, s_t = s, a_t = a \right].$$

The Bellman Optimality Equation

In an MDP, the optimal Q-value function $Q^*(s, a)$ satisfies the **Bellman Optimality Equation**:

$$Q^*(s,a) = \sum_{s'} P(s' \mid s, a) \left[R(s,a,s') + \gamma \max_{a'} Q^*(s', a') \right].$$

In expectation form the latter is given by

$$Q^*(s,a) = \mathbb{E}\left[R_{t+1} + \gamma \max_{a'} Q^*(s_{t+1}, a') \,\middle|\, s_t = s, a_t = a \right].$$

Derivation: Bellman to TD Update Rule

1. **Sample Approximation:** Model-free learning replaces sum over unknown transitions with sample transition (s, a, r, s'):

$$\text{TD Target} = r + \gamma \max_{a'} Q(s', a')$$

2. **Exponential Moving Average Update:** Update $Q(s, a)$ towards sample target with learning rate α:

$$Q(s, a) \leftarrow (1 - \alpha)\, Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') \right]$$

3. **Temporal Difference (TD) Form:** Reorganize terms to yield standard Q-update rule:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \underbrace{\left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right]}_{\text{TD Error } \delta}$$

Key Hyperparameters

Learning Rate ($\alpha \in (0, 1]$)

Determines weight of new experience vs. prior knowledge.

- **Small α:** Smooth, stable, but slow convergence.
- **Large α:** Fast updates, but sensitive to stochastic noise.

Discount Factor ($\gamma \in [0, 1]$)

Determines valuation of future vs. immediate rewards.

- $\gamma \to 0$ **(Myopic):** Focuses exclusively on immediate rewards.
- $\gamma \to 1$ **(Farsighted):** Values long-term future goals heavily.

The Exploration vs. Exploitation Trade-off

Exploration

Selecting unfamiliar actions to discover state-action spaces and hidden rewards.

Exploitation

Selecting best known action ($\arg\max_a Q(s,a)$) to maximize short-term return.

The ϵ-Greedy Policy

$$a = \begin{cases} \text{Random action } a \in A, & \text{with probability } \epsilon \quad \text{(Exploration)} \\ \arg\max_{a'} Q(s,a'), & \text{with probability } 1-\epsilon \quad \text{(Exploitation)} \end{cases}$$

Q-Learning Algorithm

Algorithm 1 Q-Learning (Off-Policy TD Control)

Require: Step size $\alpha \in (0, 1]$, small $\epsilon > 0$, discount factor $\gamma \in (0, 1]$

Initialize $Q(s, a)$ arbitrarily for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$, and $Q(\text{terminal}, \cdot) = 0$

for each episode **do**

Initialize state S

repeat

Choose A from S using a policy derived from Q (e.g., ϵ-greedy)

Take action A, observe reward R, and next state S'

$Q(S, A) \leftarrow Q(S, A) + \alpha \left[R + \gamma \max_a Q(S', a) - Q(S, A)\right]$

$S \leftarrow S'$

until S is terminal

end for

Q-Learning Algorithm

Convergence Behavior

Over repeated training episodes, Q-table entries stabilize. Extracting the optimal sequence of actions requires running greedy actions:

$$a^* = \arg\max_a Q(s, a)$$

from start state to target goal.

References

1 Richard S. Sutton, Andrew G. Barto, *"Reinforcement learning: An introduction"*, Second Edition, Cambridge: MIT press, 2020.